Discover Internet of Things

Review

A systematic review on lightweight security algorithms for a sustainable IoT infrastructure

Kamal Uddin Sarker[1]

Received: 2 September 2024 / Accepted: 15 April 2025
Published online: 25 April 2025

Abstract
Operational technology, industrial automation, advanced healthcare systems, and smart city infrastructures are common forms of IoT integrated distributed networks. Numerous IoT components require vast amounts of power for sensing, data extraction, processing, and sharing over the internet. Moreover, IoT devices are in operation with a lack of standards that poses a severe security threat. Intrusion detection systems and cryptographic algorithms are two important components in the security posture of a cyberspace. IoT applications provide real-time services, so security measures are active 24/7/365. Hence, these algorithms are also related to energy consumption. It is a study of security algorithms based on their hardware and software implementations. This review systematically searches for and selects the latest research documents, conducting an analysis of sustainability and security. It depicts the lightweight features of cryptographic algorithms; the relationship of gate density, chip area and power consumption in CMOS technology; and the importance of Machine Learning (MA) applications for IDS and cryptographic solutions in Next Generation IoT (NGIoT).

Article Highlights

- IoT components are widely utilized without standards, leading to increase security threats and significant power consumption.
- Identification of sustainable features in an IoT infrastructure without compromising security.
- Lightweight cryptographic algorithms in hardware and software level implementations can enhance the sustainability of the NGIoT platform.

Keywords NGIoT · Security · Sustainability · Machine learning · Intrusion detection system · Cryptography

1 Introduction

The Internet of Things (IoT) integrates objects and sensors that provide communication among themselves without any human intervention. IoT demands are extensively increased in automation services in smart homes, industries, smart cities, smart farming, healthcare systems, smart grids, smart traffic systems, all of which are coupled with foolproof security requirements. Billions of devices are interconnected in IoT infrastructures to aid communication

✉ Kamal Uddin Sarker, kusarkerumt@gmail.com; ksarker@ocmt.edu.om | [1]Department of Computer Science and MIS, Oman College of Management and Technology, Barka, Oman.

and computing, that generate vast amounts of data under the constraints: smaller memory, limited power, and lower processing capabilities than traditional information technology. Devices, sensors, humans, and services of an IoT platform are connected universally and continuously increased. IoT infrastructures face more challenges than traditional IT due to (i) resource constraints (power, memory, and processor) features, (ii) hostile, dynamic and heterogeneous environment, (iii) big data handling demands, (iv) real-time services, (v) lightweight applications and security algorithms, and (vi) special protocols. Figure 1 illustrates the challenges of an IoT infrastructure. This review article depicts, analyzes and compares the lightweight cryptography algorithms.

Classical IoT infrastructure is decentralized with clouds, dynamic roaming services, and multi-paradigm computing systems: serverless, NoSQL, event-sourced, microservice-based. IoT-based smart applications are increased with powerful sensors and actuators due to advancements in technologies. IoT components are embedded in industrial installations, public infrastructures or private spaces within a critical nomadic environment. IoT devices communicate through a vulnerable Wi-Fi environment to execute their typical operations such as collecting, processing, storing, sharing, and accessing information.

An imperative security design concern has become necessitated to safeguard the resources and services of any organization that demands additional resources and energy. Thousands of IoT devices are always functioning and demand a significant amount of power in a single physical environment, though it is a low power-consuming devices. Every day, numerous IoT components are integrated into the internet to enhance operational efficiency, automation and scalability services. As a result, more traffic and cyber threats have emerged. The remaining part of this section will explore the opportunities and challenges of NGIoT, identify research gaps, and formulate research questions.

1.1 Next generation IoT

Pervasive computing addresses the novelty of complex NGIoT ecosystems with 5G/6G communication technologies, heterogeneous physical entities, complex protocols, artificial intelligence (AI) applications, data science, edge computing,

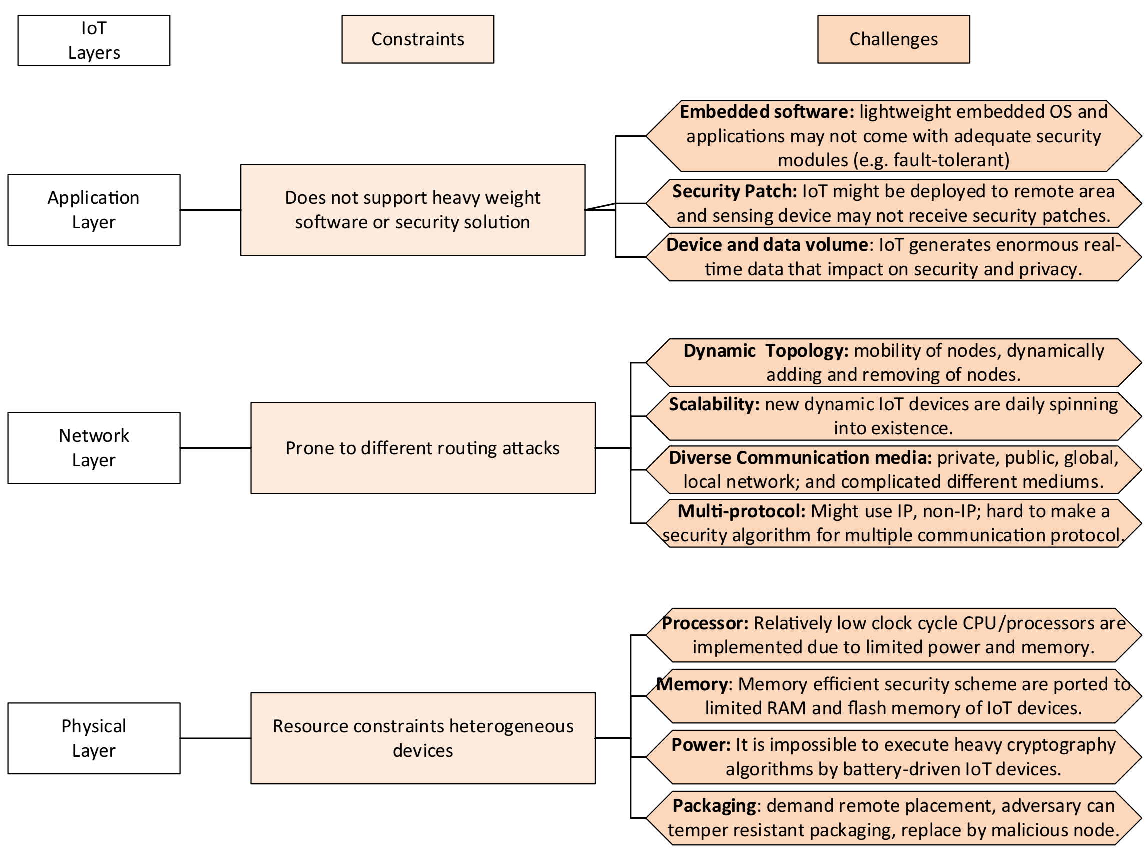


Fig. 1 Challenges of IoT infrastructure

fog computing, and cloud computing [1]. As a result, cyber physical systems experienced new types of cyber threats, more traffic, and more energy requirements. Organizations concentrate on more IoT applications to ensure quality of service (QoS) and achieve their business goals while maintaining quality factors: security, efficiency, sustainability, flexibility, portability, etc. [2, 3]. Future IoT needs to mitigate major challenges with device heterogeneity, underlying technologies, and standardization lacking [4]. NGIoT should focus on introducing sustainable hardware and ML algorithms in edge computing, fog computing and cloud computing, in addition to managing the complexity of 5G/6G technologies [5]. The novelty of this literature review relates to security algorithms and energy consumption features of an IoT component.

1.2 Sustainability formation

A sustainable action improves economic growth by minimizing production costs, reforming a healthy environment by reducing pollution, and ensuring peaceful society by demolishing discrimination and racism. These are known as economic, environmental, and social dimensions of sustainability. Sustainability practices should be initiated by the strategy of an organization and followed by implementation into processes, products, and practices. Vendors can utilize sustainable technologies to develop a smart system while an organization can promote sustainable practices by implementing appropriate processes and strategies [2]. 6Rs (Reuse, Recycle, Reduce, Repair, Remanufacture, and Redesign) is a common sustainability model that is popular in a production oriented business model [6]. A circular architecture is more sustainable than the state business architecture (take-make-dispose) [7] by implementing the 6Rs model. Our work concentrates on only maintaining optimal energy consumption of security algorithms on an IoT physical system.

1.3 Potentiality of NGIoT

The nascent stage of the IoT revolution is passing with unprecedented amounts of data processing capabilities. It has shown the tremendous impact on business outcomes that is poised to bring new dimensions of NGIoT into the global market of operational technology, industrial automation, security surveillance, supply chain management, business process optimization and real-time decision making systems.

Sustainability by NGIoT: A sustained IoT service utilizes energy efficient hardware, effective data transmission techniques, power-aware scheduling, and offloaded computation [8]. IoT applications optimize automation processes that reduce carbon emissions [9–11]. Intelligent scheduling systems and logistics delivery models promote green and sustainable practices in manufacturing industries [12]. The smart wastage management system is a green IoT application [13, 14] for sustainable societies. IoT-enabled ecosystem can enhance economic sustainability by implementing the circular economy of an organization that promotes business process optimization [15], ensures raw material reuses/recycling [16, 17], redesigning IoT solutions [18], and recycling devices/materials/products [19]. Information sharing through low power IoT infrastructure saves energy and reduces resource wastage [20]. Industries implement ML applications in an IoT system to modify the composition of raw materials that enhances the utilization of localized raw materials [21]. A significant amount of energy is required for vast numbers of IoT applications that are added to the internet [22]. This review article illustrates reduction methods of energy consumption by utilising lightweight algorithms and implementation techniques.

Sustainability technology for NGIoT: Wireless Sensor Networks (WSNs) ensure efficient and sustainable communication between sensors and hubs [23]. The Passive Radio Frequency Identifier (RFID) is developed to track IoT objects that can operate without a source of energy [24]. RFID has recycling facilities and it is usable for tag systems of IoT technology [25]. The IoT platforms can implement Internet Protocol (IP) to develop smart object tracing systems. It can track organic wastage and transfer to the respective litter of inert, odorless, and sustained wastage [26]. Efficient waste handling enhances recycling facilities and reduces carbon emissions during wastage collection and transportation [14]. These services should be managed by secure and cost-effective algorithms in resource-constrained IoT nodes. Our review study concentrates on lightweight security algorithms to minimize computational complexity and energy requirements.

1.4 Constraint of NGIoT

IoT devices have limited processing and storage capacity in addition to the constraint of battery power that affects the implementation of complex security algorithms. 32-bit ARM Cortex-M3 and 8-bit AVR are resource constraint

Table 1 IoT technology selection principles

No	Principles
a	Reduce CPU and memory utilization to deduct power, and energy can save by implementing fewer execution cycles
b	Less resource-constrained devices are preferred for software implementation
c	Optimize throughput and energy utilization by reducing processing cycles because the voltage and clock cycle are fixed by a microcontroller
d	Memory constraints negatively affects the performance of a device because more cycles are required for smaller memory element utilization
e	A small area circuit predisposes less power consumption
f	The power factors vary according to technology
g	Latency is independent of technology, power consumption, and throughput while area of a device depends on technology

Table 2 Recent surveys

Review article	Hardware		Software	Sustainable features	
	Device design	Technology selection	Lightweight cryptography	Manufacture level	Maintenance level
[102]		√	√	√	
[103]	√	√	√	√	
[75]	√	√	√	√	
[104]		√	√		√
[105]	√		√	√	
[106]			√		√

devices in the IoT platform which appeal lightweight cryptographic (LWC) algorithms [27]. The hardware and software implementation of IoT depends on the size of ROM and RAM, execution speed, environment, and energy specifications [28]. Additional consensus should be maintained when resource constraints IoT devices are integrated by different manufacturers. Anomaly based Intrusion Detection Systems (IDS) should be designed carefully so that it can execute security algorithms and protocols despite limitations such as device heterogeneity, underlying technology, lack of standardization, limited memory, and low energy of IoT objects [4, 27]. Moreover, the IoT ecosystem is not easy to change rapidly [29]. IoT infrastructure becomes more manageable concerning technical aspects, power consumption, social regulations and interoperability by embedding blockchain technology [16, 30, 31]. This review paper aids selecting a set of security components from lightweight IDS algorithms and cryptography algorithms (CAs) to improve security posture while ensuring optimal energy consumption. Table 1 presents the established principles that could help to select technology and algorithms for NGIoT, which should be upheld for a better understanding of this study.

1.5 Research gap

Table 2 illustrates current survey trends on resource constrained IoT infrastructures. Most of the surveys concentrate on efficient algorithms and technologies, but this work shows equal importance on all aspects that is specified by Table 2 headings. Sustainable features should be considered during hardware design, technology selection, software development, and deployment phases that motivates to continue this survey. Many researchers have noted the role of IoT in sustainable development [9, 10, 13, 16, 17, 19, 29, 30, 32–36] through sustainable hardware [37–44] and efficient software [2, 45–55] implementations of an IoT framework. Radio frequency, high traffic, underlying heterogeneous hardware, limited buffer and processing capabilities face challenges of an IoT network [1, 3–5]. Hardware [28, 56–64] and software [65–74] level security solutions are preventing cyberattacks on IoT infrastructures but IoT vulnerabilities are not demolished (see the daily attacks at https://www.cvedetails.com/vulnerability-list/year-2023/vulnerabilities.html). According to Thakor et al. (2021), most of the lightweight cryptography algorithms like AES, SKINNY, GIFT, PRESENT, RECTANGLE, Klein, LED, TEA, XXTEA, GOST, and KTANTAN affect one or more types of attacks of cryptanalysis, key, and/or side-channel [75]. IDS uses complex algorithms that increase computation time and power consumption. There are no remarkable works that initiate to draw an optimal line between security and sustainability. This review work is motivated to conceptualize improvement areas of security and sustainability in an IoT infrastructure. It will analyze hardware and software level security algorithms, as well as sustainability measures. An in-depth analysis is required for security algorithms and IoT technology so that we can extract the scope of sustainability improvements within the desired security posture. This study addresses the following research questions to clarify the research gaps.

a) What are the hardware and software implementations in IoT security?
b) What are the hardware and software implementations in IoT sustainability?
c) What is the consequence of security algorithms on sustainable practices of an IoT framework?

1.6 Contribution

This article focuses on the latest technologies and algorithms that can ensure sustainable and secure NGIoT infrastructure. LWC algorithms and IDS algorithms are analyzed with regards to their contributions to security and sustainability. NGIoT infrastructure can leverage ML algorithms and effective hardware technologies for IDS solutions to improve security and/or sustainability. Likewise, improved hardware and software implementations of LWC algorithms can bloster security within an optimally sustainable environment. It compares technologies for IoT technologies, LWC algorithms, and IDS security devices so that IoT infrastructure designers can select appropriate components without compromising security.

1.7 Paper organization

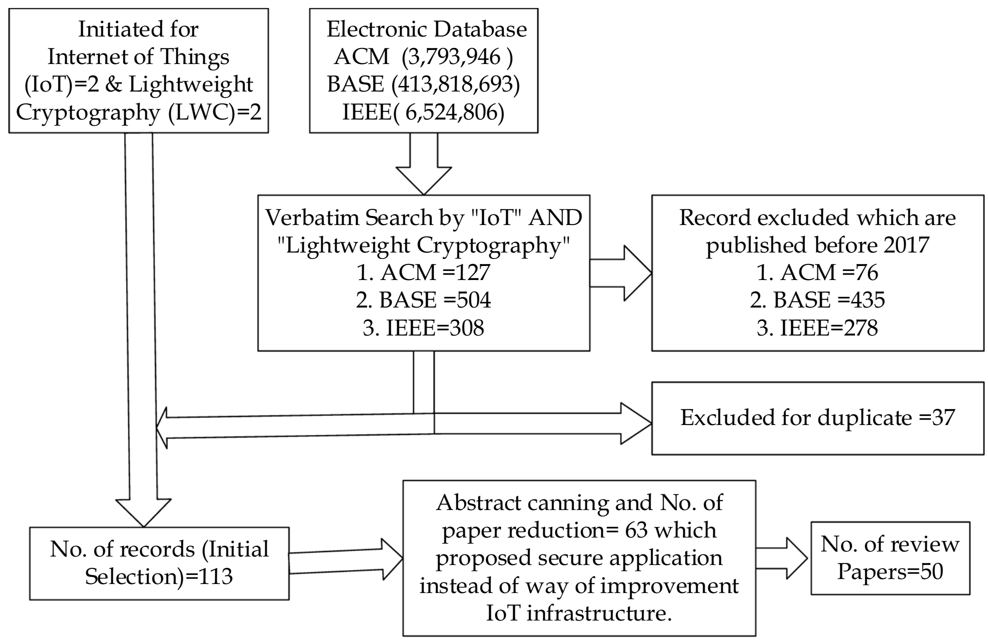
The rest of the paper is organized as follows: Sect. 2 consists of article selection, scrutiny, and finalization methods. Section 3 reviews the selected papers based on implementations (hardware or software) and their purposes: security and/or sustainability. The research is constrained by IoT-based cyber physical systems and their security algorithms (IDS and LWC). Potential sustainability issues that appear during the security study of the IoT ecosystem are reflected in Sect. 4. Finally, Sect. 5 comprehensively summarizes our findings, limitations, recommendations and concluding remarks.

2 Research method

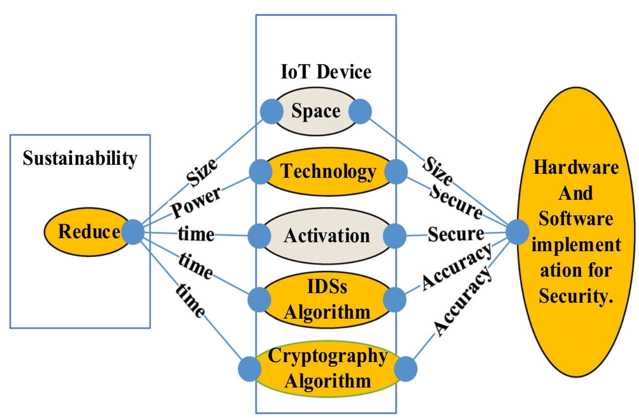
The number of IoT devices, activation time, information processing and sharing time, and protocol execution are energy-consuming factors of an IoT network [76]. An IoT-based industry generates huge electronic wastage [77] that is impossible to recycle or reuse and pollutes the environment. High investment, additional cyber threats, inadequate internet connectivity, lack of standardization, and non-degradable resource utilization are remarkable obstacles to IoT implementation [7]. The aim of this article is to explore the scope of sustainability and security improvement. So this article selects ML-based IDS algorithms for network protection and cryptography algorithms (CAs) for data encryption, where both have scope to improve in hardware and software level implementation. IoT network designers prefer less power-consuming security algorithms and technologies to improve sustainability. The following subsections illustrate the article selection process and finalization approaches used to conduct the review.

2.1 Primary searching

Google Scholar is the primary and common source of online platform that is used to search with a combination of keywords: "IoT sustainability 2022–2023", and "IoT security 2022 – 2023" because our study was started at the beginning of 2023. 2 fundamental research articles are selected from IoT and another 2 articles from lightweight cryptography algorithms which is mentioned in initiated phase of PRISMA diagram (Fig. 2a). After analyzing these articles, documentation searches were initiated for IoT and LWC following the PRISMA guidelines. The documentation search strategy is presented in Fig. 2a with ACM digital library, BASE and IEEE being the three main repositories for selecting and reviewing articles. Though the primary keywords are “IoT” and “Cryptography” for article selection, it includes more articles for an in-depth understanding of technology, hardware, and software specifications. A significant amount of research has been published in recent years which has proposed a secured and sustainable IoT infrastructure. For example, an IoT-based industry can monitor and track an item to suggest recycling a product [13], or its components [32], products’ package [33], digital product [29], and simulate a robot [19]. An IoT platform can reduce wastage of non-performing products [30], cost minimization [34], better production prediction [10], inventory



(a) PRISMA review diagram



Quality Factors	NGIoT	
	Hardware	Software
Security	10	10
Sustainability	8	12

(b) (c)

(b) Research domain and 2(c) heat map: number of articles selection for core study

Fig. 2 **a** PRISMA review diagram. **b** Research domain and **c** heat map: number of articles selection for core study

cost reduction [35], as well as good estimation of demand, production and consumption [36]. IoT architecture is used to monitor packaging [33] or robotic application manufacturing [19] from simulated output. IoT feedback systems support product redesigning [17], especially digital products [29] and robotics [19]. These papers help us to know how IoT improves security and sustainability.

2.2 Document selection

The aforementioned articles (Sect. 2.1) support detecting exact points of security domains like IDS and CAs including important research areas: lightweight cryptographic (LWC) algorithms, IoT and resource constraint devices, security algorithms and power/energy consumption. Preliminary searching is used to fix the principles that are mentioned in Table 1.

So I prefer recent articles which consist of information relates to "IoT technology", "IoT security" "IDS algorithms for IoT", "cryptographic algorithms for IoT", and "power or energy consumption by IoT devices".

Approximately 50 articles were selected by scanning abstracts, keywords, and conclusions based on the highlighted areas of Fig. 1a (amber) between 2017 and 2024. Journal articles, conference papers and books are accepted as research documents for this literature review. Secondly, I focused on the objectives of the study in 4 sections that are visualized by the heat map (Fig. 2b). Finally, 20 papers for security (hardware implementation [28, 56–64], software implementation [65–74]) and 20 papers for sustainability (hardware implementation [37–44], software implementation [2, 45–55]) were finalized for core analysis from the selected 50 articles by PRISMA. The remaining 10 articles which are selected by PRISMA, web links, and books support the initiation, development, and verification of the completeness of this research.

3 Review

This section consists of core review analysis of IDS algorithms and CAs with underlying technologies and their implementation (Fig. 2b). The breakdown is differentiated between hardware and software level implementations which is founded on 40 scholarly documents (Fig. 2c).

3.1 IDS algorithm

IDS is a device that protects a network and its components from unexpected malware data packets or denial of service (DoS) data packets. An ordinary IDS uses malware images (databases) to protect a system from known malware, while an anomaly-based IDS is used to protect the system against zero-day attacks (unknown malware) by implementing ML applications. Its throughput highly depends on the technology of the hardware platform and the effectiveness of real-time applications [28]. Table 3 represents a comparative study on hardware level IDS implementation.

Hardware based IDS has become popular with FPGA technology [56], which is extended as HH-NIDS [28] by utilizing hardware accelerators (reprogrammed) for improving security and sustainability. A traditional hardware-based IDS is implemented on heterogeneous platforms like microcontroller, SoC FPGA, and GPU [28]. Different types of datasets are used to verify their accuracy (Fig. 3a) where the IoT-23 dataset is more usable and reliable [78]. 15 papers (Fig. 3a) provide accuracy of their IDS algorithms as a quantitative data (percentage) from our selected paper list during 2019–2022. Figure 3a depicts the progress of IDS by implementing ML and deep learning (DL) algorithms. The Y axis of Fig. 3a represents percentage accuracy, where the IoT-23 dataset [79, 80] shows better accuracy (yellow) even when it combines with other datasets [69, 71, 72, 81] (amber); the accuracy is followed by 'self-generated' [70, 82, 83] and 'others dataset' [57, 70, 84–87]. Figure 3b illustrates 4 distinct resources (LTU, LTURAM, FF, BRAM, DSP) on the X axis and their percentage of hardware utilization on the Y axis based on the examined results of a research [28]. Each resource is applied to 2 approaches (HLS, Verilog) and 2 datasets (IoT-23, UNSW-NB15) on the PYNQ-Z2 SoC FPGA platform. When DSP is utilized to the maximum for any approach or dataset, respectively LUTRAM is in the lowest in utilization category.

Table 3 shows that different types of Deep Learning (DL) algorithms are implemented with FPGA and other technologies for hardware level IDS implementation of an IoT framework. But IDS algorithms do not concentrate on implementing power-saving features. When it is implemented on the perimeter of the network, it is a single machine execution. But at host level implementation, it will be a subject of energy consumption and sustainability issue. Table 4 performs a comparison analysis on recently published software level implementation on the IDS of IoT networks. Software level implementation also has the influence of DL algorithms, but some articles concentrate on implementing power-saving actions besides ensuring security.

3.2 Cryptographic algorithms

Cryptography provides tools to protect data, secure communication, and establish trust, which is known as a cornerstone of modern cybersecurity. Key management and quantum computing threats increase limitations and challenges, in addition to the resource-constrained IoT devices. A robust and secure digital channel strikes a balance between cryptography's power and mitigating its limitations. Nowadays, researchers are addressing the paradox of sustainability and security to achieve an optimal solution.

Cryptographic Algorithms (CAs) are executed on IoT nodes, central machines and networking devices that demand vast amounts of energy. Renewable energy, effective charging technology, smart routing, and ML applications are remarkable sustainable practices in IoT infrastructure [8]. To address and resolve this issue, CAs are updated with

Table 3 Hardware level IDS implementation

Refs.	Years	Contribution	Significant of the study	Limitations
[28]	2023	Heterogeneous HW-based IDS	Good accuracy is noticed for microcontroller and SoC FPGA	Not tested by FPGA network specialized hardware
[56]	2021	FPGA HW-based IDS	Good accuracy for Xilinx FPGA System on a Chip (SoC)	Offloading modules of FPGA are challenged
[57]	2020	Hierarchy IDS architecture	Reduce false positive by using MultiModal Deep AutoEncoder	Limited by general testing scenario
[58]	2017	Hardware intrinsic security	Next generation microcontrollers and SoC/ are explained	No implementation evidence
[59]	2018	FPGA-based Security system	Accelerating log processing and monitoring system	Avoid the possibility of subtle intrusions
[60]	2020	Microcontroller in IoT infrastructure	Microcontroller applications for security in IoT are reviewed	Ways of implementation is not guided by the study
[61]	2020	FPGA-based security monitoring	Flexible, adaptable, and tunable Modular Semantic Caching Framework is proposed	Offloading modules or services to FPGA are challenged
[62]	2021	Deploying CNNs on FPGAs	Reviewed challenges and limitations	Only theoretical background without critical evaluation
[63]	2019	Methodology to implement FPGA model	Aimed to reduce the latency of the FPGA applications in IoT	It is only evaluated for single dense layer
[64]	2019	FPGA with NN in IDS	A flexible model is proposed with FPGA SoC	Limited performance is observed on Zynq Arm core

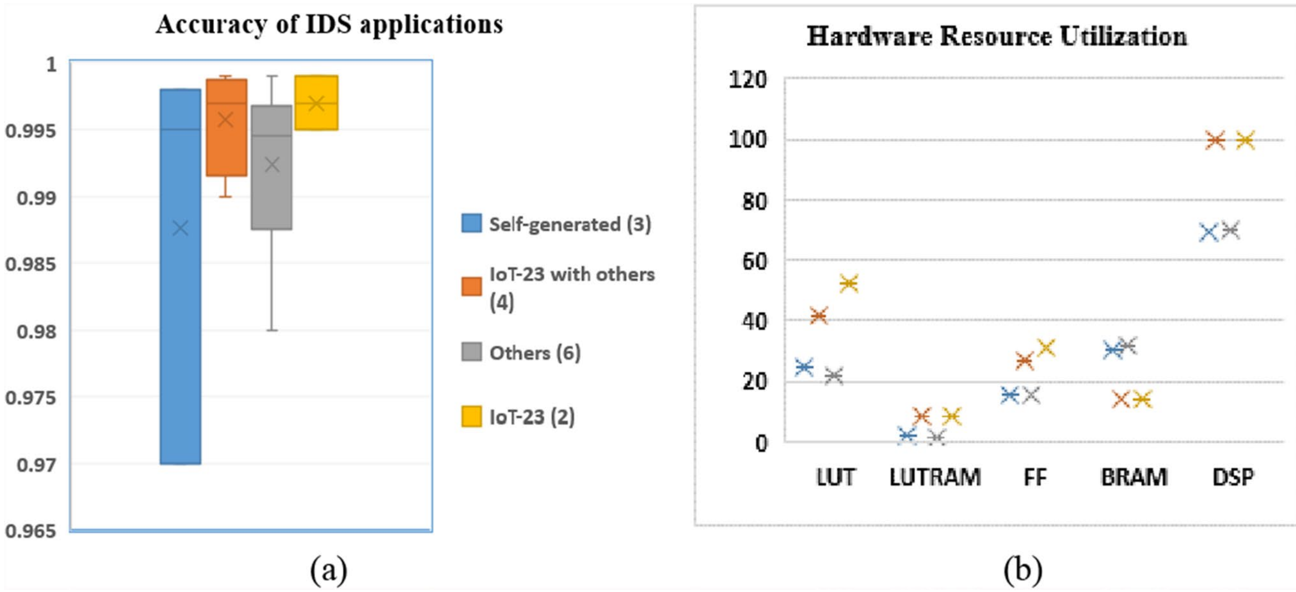


Fig. 3 **a** Hardware based IDS algorithms' accuracy and **b** hardware utilization by IoT technologies

respect to hardware and software level implementations. Figure 4 illustrates the classification of CAs on account of sustainability.

Among the three categories, the lightweight cryptographic (LWC) algorithm is preferred because it is the most sustainable option that can be implanted in various technologies. The LWC algorithms are as follows:

Lightweight block cipher (LWBC): LWBC is a symmetric cipher that is used in hash functions and message authentication codes [88]. It is structured on a substitution-permutation network (SPN) and Feistel round functions that supports minimum overhead for encryption and decryption by using the same program code [52]. As a result, it is applicable to hardware with low average power and low memory usage. The influential parameters are structure, rounds, key size, and block size of the LWBC algorithms. A notable study [52] presents a comparative study of 21 LWBC algorithms which were proposed from 1994 to 2018 where Feistel structure is mostly used. Their study also showed the existing weaknesses of the ciphers. SNF is the latest LWBC based on Feistel and SPN structure, which uses 96-bit and 64-bit keys and block sizes respectively, that completes 32 rounds in the entire encryption process. Till now there is no vulnerability record of NNF LWBC. The lightweight block cipher is commonly applicable in RFID technology and sensor networking applications such as parking management, object identification, and goods tracking.

Lightweight stream cipher (LWSC): LWSC is implemented using CMOS technology alongside various types of registers. It encrypts and decrypts 'r' bits simultaneously, with its parameters being key size, chip area, Initialization Vector (IV), Internal State (IS), and throughput. Dhanda S. S. et al. [52] depicted a comparative analysis of 20 LWSC algorithms from 1987 to 2017 and found that Linear Feedback Shift Register (LSFR) is used most frequently. The latest LWSC (lizard) is a Non-Linear Feedback Shift Register (NLFSR) tool with a 120 bit key size and 121 internal states which has a low-cost implementation [52]. Lizard occupies a 64 bit initial vector and 1161 GE [89]. Besides this, Espresso [90] is designed for 5G applications that is founded on the NLFSR register structure. LWSC is good for real-time communication applications where an optimal situation is preferred between performance and security.

Lightweight hash function (LWHF): The hash function is used to ensure data integrity during transmission that generates a fixed length hash value from an arbitrary-length message. Neeva is considered the latest LWHF, which is founded on the sponge structure and faster than Spongent-224 [91]. It provides 112 bits of collision resistance that can be used for RFID technology. Hash-One, L-Hash, GLUON, and SPONGENT are famous hashing algorithms used for various applications and founded on different technologies including IoT based blockchain [52].

Elliptic curve cryptography (ECC): The asymmetric system utilizes larger keys and higher memory which increases costs. RSA and ECC are primitives that reduce such problems in public key cryptosystems, but ECC ensures the same security level as RSA with a comparatively smaller key size [52]. Till now, ECC is popular as the choice for IoT security [52], although it is 100–1000 times slower than AES on an 8-bit microprocessor [92]. It is also implemented in blockchain technology and digital signatures.

Table 4 Software level security

Refs.	Years	Contribution	Significant of the study	Limitation
[65]	2022	Anomaly based IDS	Application of AI algorithms in cyber security is reviewed	No technology specified dependency
[66]	2019	Anomaly based IDS	AI and ML algorithm for IoT sensors	Analysis depends on only five algorithms
[67]	2021	Anomaly based IDS	Application of DL in IoT security is reviewed	Explore only 7 algorithm without any model
[68]	2021	Anomaly based IDS	Introduced fog computing to improve performance of IDS	Only three algorithms are incorporated to test
[69]	2019	Intelligence attack analysis	Deep Neural Network to classify unseen and unpredicted attacks	No monitoring and detection prior to attack
[70]	2021	Distributed IDS	Graph NN algorithm for monitoring IoT devices, SDN forwarders, Fog Nodes	Highly influenced by nearly agents
[71]	2022	Anomaly based IDS	Light weight Recurrent NN for IoT infrastructure	Optimization technique is not implemented
[72]	2020	Anomaly and attack detection	Ensemble an new method that leverages deep models	State-of-the-art without implementation
[73]	2022	Malware detection model	Convolution NN showed better performance	High processing and consumption
[74]	2022	Anomaly detection model	Improve performance through gradient boosting (GB) and decision tree (DT)	Limited by open-source Catboost for IoT security

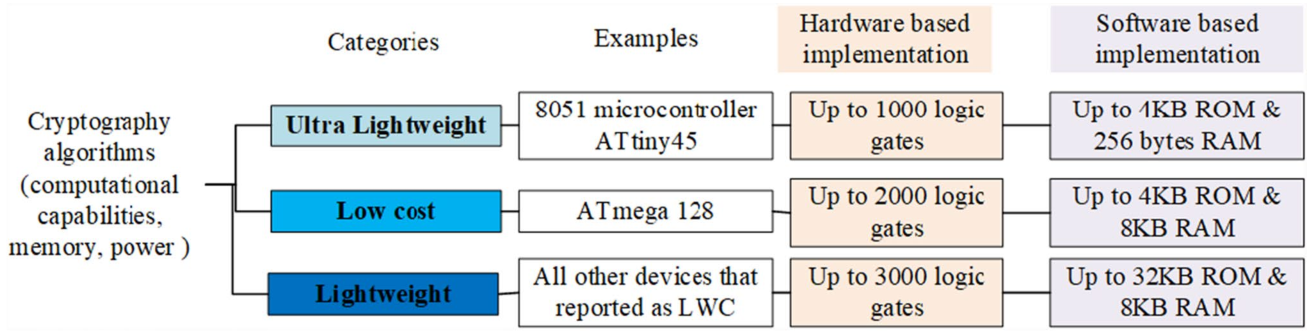


Fig. 4 Classification of CAs for sustainability

3.2.1 Hardware level implementation of LWC algorithms

IoT technology has historically improved energy saving features through the 802.11a, 802.11b, and 802.11 g standards for short-range and low bandwidth communication. Meanwhile, 802.11ac, 802.11ad, and 802.11n are utilized for short-range and high-bandwidth communication [37]. Low-Powered Wide-Area Network (LPWAN) technologies offer wider coverage with restricted power consumption [37]. Cellular networks can be utilized by LPWAN for IoT services to improve reliability, security, energy efficiency, and coverage area though it has limited data transmission speeds [38]. SigFox [39] is a low power-consuming solution that connects IoT devices and sensors to save energy and improve autonomy. It is cost-effective and simple but can transmit a message of up to 12 bytes at a time only. Narrowband IoT (NB-IoT) is commonly used in smart city applications such as parking monitoring, object tracking, and healthcare applications. It is a lower power-consuming and higher speed technology than LPWAN, but it is expensive, complex and has lower bandwidth [40]. Long-Range Wide Area Network (LoRaWAN) is a bidirectional, low-power consuming communication technology for IoT that has a communication range of up to 15 km but the maximum data rate is 27kbps [41]. The longevity of a battery depends on the physical layer data transmission technology [42], utilization of energy saving parameters like: power saving activation mode, device sleep mode, etc. [43] and low power-consuming communication techniques [44].

3.2.2 Software level implementation of LWC algorithms

Energy consumption of an application depends on computer language, data structure, algorithm, architecture, loop, logic, and coding utilization [2, 45, 46]. Implementing ML APIs for web data collection [47], and nano data centers save more energy than traditional approaches [48]. Resource virtualization, lightweight applications, and ML apps are some techniques to reduce power consumption [49]. CAs are complex, requiring more CPU burst time, and they face difficulties in the resource constrained environment (limited computation, memory, size, and power) of IoT [50]. LWC algorithms contribute to solving technical issues, ensuring security, and reducing power consumption [50] in a resource-constrained environment. Noticeable energy requirements vary with IoT protocols [51], but a little bit energy consumption is varied for security requirement updates. Device specifications (technologies, cores, memories, CPU clock speeds, etc.) and protocols are influential factors to measure the performance and energy consumption along with other factors (key size, round times, computation complexity, etc.) of a security algorithm [51]. The researchers divide LWC primitives into four elements: LWBC, LWSC, LWHF, and ECE (details at Sect. 3.2) [52]. On the other hand, the implementation of the LWC is categorized into three types [55], which is depicted in Fig. 3. Ultra lightweight ciphers include KATAN, KATANTAN, QT1, Fruitv2, Piccolo, Sprout, and HUMMINGBIRD; low-cost algorithms are PRESENT, MIBS, SIMON, TWINE etc. and LWC algorithms are CLEFIA, DEXI, SOSEMANUK, and ECC [52]. A comprehensive analysis of LWC algorithms is performed by two distinguished recent works [53, 54] for resource constrained devices with respect to the aforementioned four ciphers. This review work will compare a set of LWC algorithms (AEX, PRESENT, SEA, TEA, KATAN, SIMON, Klein, and Camellia) that are commonly analyzed by NIST concerning throughput, efficiency, and required power, in addition to comparing sustainable practices in IDS and cryptography.3.2.3 Cryptography Libraries for IoT.

Numerous cryptography libraries have been integrated into a multitude of encryption algorithms that enable various security measures. In this section, a few libraries are portrayed that are suitable for IoT infrastructure.

- *WolfSS:* It is designed for use in embedded systems due to its small footprint size and low runtime memory requirements (1–36 KB). It supports developing cross-platform algorithms under the General Public License GPLv2. It includes an OpenSSL compatibility layer to validate certificates. This library is applicable for a resource constrained environments like a small APIs, and it supports zlib for compression in IPv4/IPv6 networks.
- *WiseLib:* It is used in embedded networking devices so that an individual can compile algorithms for various platforms like Shawn, Contiki, and jSense. It uses elliptic curve cryptography over prime fields, and it is applied to in-house routing/localization algorithms. WiseLib templates facilitate compilation of the platform independent code, but this shuns assembly level optimizations.
- *AvrCryptoLib:* It uses C-interfaces to perform modular exponentiation of an 8-bit assembly language in an AVR 8-bit microcontroller to reduce execution time in its cryptosystem. It facilitates MAC functions and PRNGs (Pseudo Random Number Generators) for cryptographic applications. It improves the efficiency of SRAM consumption by storing keys in flash memory.
- *RelicToolKit:* This is a flexible meta-toolkit for constructing customized cryptographic toolkits. It supports easy compilation by including multiple-precision integer arithmetic. It is used for component building and inclusion for individual platforms. It is highly recommended to develop an independent toolkit for the optimum performance of a desired platform.
- *TinyECC:* It is a library that is founded on elliptic curve cryptography, which is utilized for primary key cryptographic operations. It can be integrated on different other devices by modifying code, although it is specifically designed for TinyOS dependent devices. It is applicable in security algorithms for verification in sliding windows, and Barrett reduction. The prominent features of TinyECC are digital signatures, public key encryption, and key exchange protocol.

4 Fact analysis of IDS and LWC

This section summarizes the outcomes of the review articles that could contribute to enhancing sustainable features in the security algorithms. Though these algorithms were invented for security, nowadays the latest versions are designed for resource constrained devices.

Advancements of hardware technologies enhance computational capabilities, reduce the size and energy requirements of the devices. Figure 5 illustrates the specifications of security algorithms based on 4 domains of this research according to Fig. 2. FPGA technology is commonly integrated with other technologies to increase growth, variation and throughput for hardware implementation of IDS. It does not indicate the importance of reducing power-consumption; hence it is better to implement at the perimeter of IoT networks. DL algorithms like neural networks and convolutional neural networks are commonly used to enhance the accuracy of IDS algorithms. The software level implementation of IDS uses modified DL algorithms, which are known as lightweight algorithms that can function on limited resources like RAM, energy, computational capability, and physical area of the device. Many LWC algorithms are implemented at the hardware level as well as at the software level. The implementation of low-power technology, low-power communication mediums, optimal activation time management applications, and more energy-efficient algorithms are the main methods of hardware level implementation of LWC. Software level implementation works on different lightweight cipher algorithms beside traditional CAs as well as energy efficient techniques such as implementing a nano data center instead of a centralized data center and accepting web data instead of sensor data if data consistency is not violated.

Designers always expect the ideal implementation of a lightweight cipher that should consume the least amount of power, utilize minimal resources, occupy a small circuit area, and perform secure communication as fast as possible. Figure 6 shows a comparison study of gate density and power consumption with respect to CMOS technology [55]. It is clearly reflected that power consumption highly depends on the size of a chip. Table 5 highlights the enhancement areas for LWC algorithms where research can focus on any primitive separate to design a new algorithm.

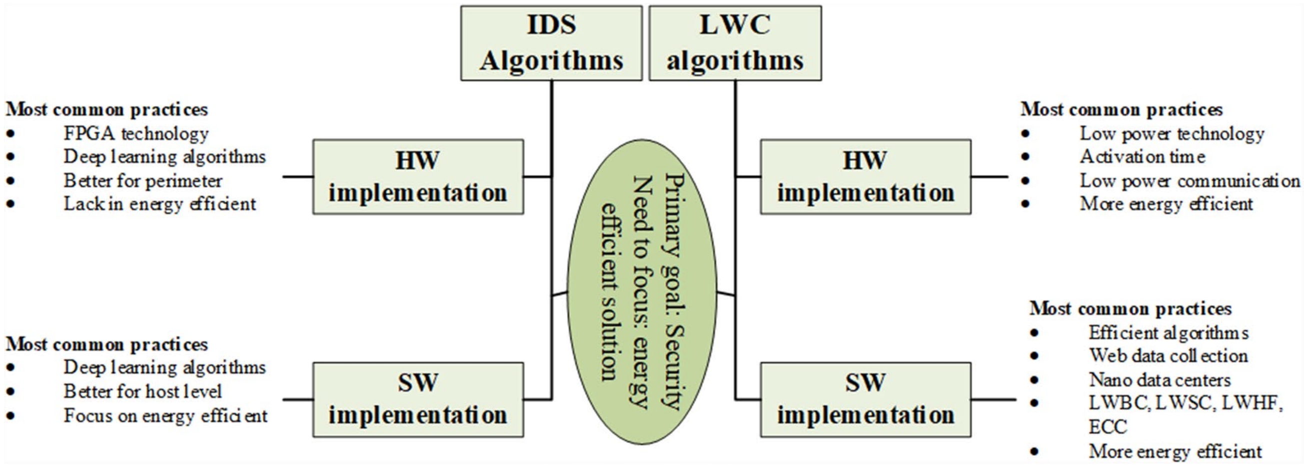


Fig. 5 Roles of IDS and CAs in IoT ecosystem

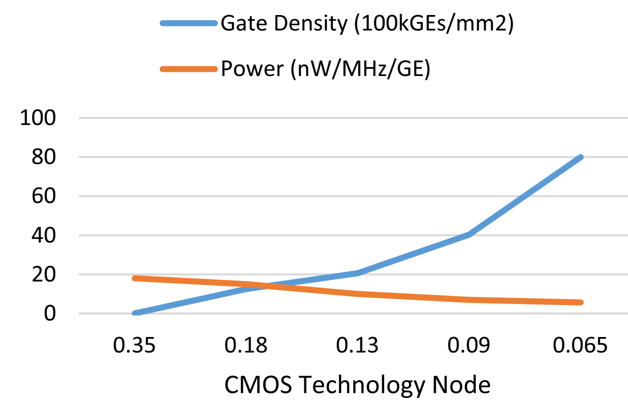


Fig. 6 CMOS technology nodes for commercial standard-cell libraries [55]

5 Conclusion

Highly distributed systems, big data handling, numerous physical objects and sensors, lightweight applications, and chip areas are influential factors in increasing the challenges like security, latency, and energy consumption [52]. A system cannot maximize its throughput without a better adjustment of all parameters; for example, a high latency system could be outdated and ineffective for real-time applications. Table 6 explains the outcomes of the study according to the heat map (Fig. 2b).

Security is the primary factor of a cipher that is evaluated for its resistance against cyberattacks. The length of a key represents the strength of a symmetric cipher, while the half-length of the key represents the strength of an asymmetric cipher. Lesamanta-LW (128), and SPONGENT (128) are the most secure in LWHF [52] to provide data integrity.

The chip area is the hardware implementation and importance for sustainability in aspects of power and cost that is measured by gate equivalence (GE). It directly affects GE and energy consumption, so an IoT device with a smaller cross sectional area is always preferable [54, 55]. It is the ratio of the layout area of the application (µm2) and NAND2 gate area [55] that is highly dependent on technology (CMOS). If technology changes from 35 to 0.065 µm, the gate density becomes 800 (kGE/mm^2) from 6800 (kGE/mm^2) as well as power consumption decreased from 18 (nW/MHzh/GE) to 5.68 (nW/MHz/GE) [55].

Throughput is used to determine the efficiency of a cipher implementation (hardware or software based) that should be as high as possible for a hardware implementation. According to Dhanda S.S. et al. [52], the highest throughputs of LWSC, LWBC and LWHF algorithms are respectively Grain-128, PRINCE, and RECTANGLE respectively.

Table 5 Area of enhancement for ciphers

Algorithms	Scope to improvement
LWBC	Reduction block size and key size without compromising security: A block size of 128 bits or larger is accepted in modern block ciphers, which can ensure a sufficient key space and improve the security of the block cipher. However, this increases computational complexity and makes it impossible to execute on a resource constrained device. Hence, an IoT environment demands a secure block cipher with a minimal block and key sizes
	Implement a simple round function and reuse it: The computational complexity of a round function is determined by the size and number of S-boxes in the confusion function as well as the length of bits for the permutation process in a diffusion function. Resource constrained devices demand a simple round function with a minimum number of rounds and S-boxes without compromising security. Moreover, reusing of a round function can be an efficient approach
	Implement a simple key schedule: The key schedule generates all round keys from the key of an encryption process that ensures strength against the differential or linear cryptanalysis. A well-structured and complex key schedule is faster with a uniform distribution. In a resource constrained environment, well-designed and simple schedules are preferable due to computational limitations
LWSC	Reduce the number of keys or IV setup cycle: A 128 bit or larger block size is accepted in modern block ciphers, which can ensure a sufficient key space and improve the security of the block cipher, but computational complexity is increased, and makes it impossible to execute on a resource constrained device. Hence, an IoT environment demands a secure block cipher with a minimum block and key sizes
	Compress the cross section area of a chip: A stream cipher with a smaller area allows a smaller state size, improving the resistance against time-memory-data trade-off (TMDTO) attacks. A stream cipher that can be operated on significantly less area will reduce energy consumption
	Reduce key length: In cryptography, the key length is pivotal in computational theory and intertwines with computational complexity because a longer key can fortify the system with increased complexity. However, in a resource-constrained system, we should limit key's length to ensure sufficient security
	Internal state minimization: Lizard [52] is the newest lightweight stream cipher referred to as a low-cost implementation that utilizes Non-Linear Feedback Shift Register (NLFSR) with a key length of 120 bits and 121 internal states. The number of internal states is one of the crucial aspects in the computational complexity of a stream cipher algorithm and research may aid in enhancing the algorithm by diminishing the number of internal states without compromising security
LWHF	Minimize size of message and output: A hash function digests an input message and generates a fixed length output by utilizing a one-way function. Lighter encryption and authentication schemes are necessary to ensure data integrity for IoT devices and a lightweight hashing function is the most common one. A bigger input message is a computationally complex and lengthy process. A longer output is comparatively difficult for cryptanalysts, which is preferred for security, but a resource constrained environment suggests as little as possible without compromising security
ECC	Speedup execution: The speedup of elliptic curve cryptography depends on the architecture of the model, underlying technology, and software implementation. Modular reduction in architecture, better technology utilization (e.g. FPGA), and AI implementation to identify a prime order for software level improvement are a few areas for improvement
	Optimization for Prime Field (PF): The elliptic curve utilizes a set of points on it that derive class variation in cryptosystems like Diffie-Hellman and DSA. The points are defined over a PF and special primes are selected to minimize modular reduction. An implementer can reduce the computational complexity of ECC by utilizing optimal prime-factors
	Reduce memory utilization: P-256, P-384 and P-521 are NIST standardized elliptic curves that could be modified for a scalable and memory-efficient implementation. Efficient algorithms, hardware multipliers, multi-precision squaring, and modular reduction are common methods to save memory
	Reduce energy consumption: The choice of technology is an important factor for hardware level implementation to save energy. Redesigning the hardware architecture by eliminating extra circuitry is a fundamental approach to reduce power consumption and speed up computation

Latency (computation time for one block of plaintext (decryption) or cipher text (encryption) is important, like security for real-time IoT solutions. LWBCs are preferable in terms of latency and the PRINCE cipher is specially designed for low latency solutions [52].

Low power-consuming technologies, applications, methods, processes, and communication techniques can reduce a significant amount of energy and power consumption, but demand for electricity will increase due to the rapidly growing IoT connectivity. Therefore, utilizing renewable energy is another sustainable action in the IoT framework.

Table 7 compares throughput, efficiency, and required power for a set of LWC algorithms on both hardware and software implementations for 8-bit microcontrollers [75]. For example, AES-128/128 (0.13) stands for the name of the algorithm-key size/block size (technology length in micrometers). Efficiency (Kbps/KGE), energy (J/bit), and throughput (kbps) are compared for both hardware and software-level implementations. IoT infrastructure designers can utilize the comparison study by Tahkor et al. [75] to select appropriate LWC algorithms and customize them.

Sustainable physical space selection, utilization of sustainable technology, sustainable protocol implementation, smart service level frequency, and real-time solutions are the most important items in IoT infrastructure [93]. The study

Table 6 Improvement requirements for NGIoT

Factors		NGIoT	
		Hardware level implementation	Software level implementation
Security	IDS	• Implement Image-based IDS and anomaly-based IDS in a series at the network perimeter • FPGA based security monitoring systems	• Implement IDS applications for all nodes to protect the device • Implement effective ML/DL applications for IDS
	Cryptography	• Select technology based on the coverage and bandwidth • Implement low latency technology to enhance security	• Deploy appropriate LWC primitives from LWBC, LWSC, LWHF, and ECE
Sustainability	IDS	• Apply master–slave technique and power-saving technique during IDS implementation • Maximize utilization by DSP technology of IDS	• Minimize as much as possible uses of IDS application if not required • Deploy comparatively lightweight ML/DL applications for IDS
	Cryptography	• Implement low power consuming technology to deploy LWC algorithms based on communication range	• Implement ML LWC algorithms • Power saving techniques like sleep mode, deactivate mode

Table 7 Performance Comparison [75]

LWC algorithm-key/block(Tech µm)	Implementation (Hardware)			Implementation (Software)		
	Efficiency (Kbps/KGE)	Energy (µJ/bit)	Throughput (kbps)	Efficiency (Kbps/KGE)	Energy (µJ/bit)	Throughput (kbps)
AES-128/128(0.13)	23.6	42.38	65.64	132.9	16.7	122
PRESENT-80/64(0.18)	127.38	11.7	200	35.91	43.1	23.7
Klein-64/64(0.18)	25.32	59.18	30.9	10.84	10.6	32.4
TEA-128/64(0.18)	42.46	35.32	100	53.24	30.3	35.5
Camellia-128/128(0.18)	44.45	33.57	290.1	6.34	256	8
SEA-96/8(0.13)	0.89	1117.67	2.29	21.6	173.7	9.2
KATAN-80/32(0.13)	15.58	64.16	12.5	10.35	289.2	3.5
SIMON-96/48(0.13)	20.7	48.32	15.8	1900	2.3	323

considers all elements except sustainable practices in the cyber physics system of an IoT environment which is beyond the scope of this study (Table 8).

Limitation: The research is still in its infancy with limited security analysis on LWC and IDS algorithms as well as low-power consuming technologies. The higher growth rate of IoT subscribers accelerates IoT devices in the public networks introducing new challenges and requiring more technical study based on the latest upcoming technology for NGIoT. It has failed to demonstrate the mathematical relationship between throughput, security, latency, and the cross sectional area of an IoT chip. This is a generalized study that is not based on any vendor's product but it shows the importance of standardization to maintain an optimal outcome [94].

Remote area IoT networks are wireless sensor networks where nodes are primarily battery operated. These are also known as battery-life constrained devices. Power management is crucial to maximize the life of a sensing device as well as sustainable practice. Carbon emissions rate, battery lifetime, battery materials, and disposal/recycling are the general concerns about battery sustainability. Recently, lithium-ion battery technology has been considered ethically and environmentally problematic [95]. Lithium-ion batteries are commercially favored for lower costs (raw materials are readily available) and high-efficiency (fewer carbon footprints) technologies [96]. However, more energy can be packed into a smaller space with solid-state battery technology [96]. Moreover, it is safer and has a relatively shorter charging time. A remote IoT network can only be sustainable when sustainable batteries are incorporated into the infrastructure.

Future work: It has explored the lightweight algorithms for hardware and software implementation which are commonly used to improve security and energy-consumption within a resource-constrained IoT infrastructure. With the massive data connectivity, large-scale sensors and actuators of an NGIoT environment will demand potential solutions with low cost, less memory, low latency, small chip area, less energy consumption, and more security. Table 7 will assist in selecting LWC algorithms that are limited to 8-bit microcontrollers only; moreover, this study does not mention other important parameters, like required area and power. Additionally, it does not discuss algorithms with their security analyses: linear /differential/integral/square cryptanalysis, key attacks, or side-channel attacks. My upcoming work will develop i) optimal requirements guidelines and ii) a mathematical relationship-model of the aforementioned IoT parameters based on NIST cryptographic standards [97]. Researchers recently proposed two cryptographic algorithms [98, 99] that are more suitable regarding time complexity and memory utilization, which could be updated for lightweight IoT infrastructure. On the other hand, IoT devices, nodes, and infrastructure management also play a significant role in sustainability practices [100, 101], and this paper will be extended to focus on advanced AI-based management

Table 8 Sustainable validity check

Domain of the study	Sustainable physical space selection	Utilization of sustainable technology	Sustainable protocol implementation	Smart service level frequency	Smart service level frequency	Real time solution
IDS	NA	√	√	√	√	√
Cryptography	NA	√	√	√	√	√

technologies. Moreover, the comparison study will be illustrated by the pseudocodes of the cryptographic algorithms; it may include sustainable batteries, technologies, and sustainable network design.

Author contributions Single author and all contributions belong to correspondence author.

Funding No funds, grants, or other support was received for conducting this study.

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

References

1. Szmeja P, Fornés-Leal A, Lacalle I, Palau CE, Ganzha M, Pawłowski W, Paprzycki M, Schabbink J. ASSIST-IoT: a modular implementation of a reference architecture for the next generation internet of things. Electronics. 2023;12:854. https://doi.org/10.3390/electronics12040854.
2. Sarker KU, Deraman AB, Hasan R, Abbas A. SQ-framework for improving sustainability and quality into software product and process. Int J Adv Comput Sci Appl (IJACSA). 2020;11(9):69–78. https://doi.org/10.14569/IJACSA.2020.0110909.
3. Sarker KU, Deraman AB, Hasan R. Ontological practice for software quality control. Int J Bus Inf Sys. 2020;34(3):355–72. https://doi.org/10.1504/IJBIS2020.108650.
4. Ali O, Ishak MK, Bhatti MKL, Khan I, Kim KI. A Comprehensive review of internet of things: technology stack, middlewares, and fog/edge computing interface. Sensors. 2022;22:995.
5. Asheralieva A, Niyato D. Optimizing age of information and security of the next-generation internet of everything systems. IEEE Internet Things J. 2022;9:20331–51.
6. Sihvonen S, Ritola T. Conceptualizing ReX for aggregating end-of-life strategies in product development. Procedia CIRP. 2015;29:639–44. https://doi.org/10.1016/j.procir.2015.01.026.
7. Ding S, Tukker A, Ward H. Opportunities and risks of internet of things (IoT) technologies for circular business models: a literature review. J Environ Manage. 2023;336:117662.
8. Alsharif MH, Jahid A, Kelechi AH, Kannadasan R. Green IoT: a review and future research directions. Symmetry. 2023;15:757. https://doi.org/10.3390/sym15030757.
9. Chit TW, Ning L, Paliath NA, Long YM, Akhtar H, Shanshan Y. IIoTenabled and data-driven sustainability evaluation framework for textile supply chain. 2021 IEEE 16th Conference on Industrial Electronics and Applications (ICIEA). 2021; 297–304. https://doi.org/10.1109/ICIEA51954.2021.9516314.
10. Ghoreishi M, Happonen A. The case of fabric and textile industry: the emerging role of digitalization, internet-of-things and industry 4.0 for circularity. Proc Sixth Int Congr Inf Commun Technol. 2022;216:189–200. https://doi.org/10.1007/978-981-16-1781-2_18.
11. Jagtap S, Garcia-Garcia G, Rahimifard S. Optimisation of the resource efficiency of food manufacturing via the Internet of Things. Comput Ind. 2021;127:103397. https://doi.org/10.1016/j.compind.2021.103397.
12. Liao W, Wang T. A novel collaborative optimization model for job shop production–delivery considering time window and carbon emission. Sustainability. 2019;11(10):2781. https://doi.org/10.3390/su11102781.
13. Al-Masri E, Diabate I, Jain R, Lam MH, Nathala SR. Recycle.io: an IoTenabled framework for urban waste management. In: 2018 IEEE International Conference on Big Data. IEEE. 2018; 5285–5287. https://doi.org/10.1109/BigData.2018.8622117.
14. Velvizhi G, Shanthakumar S, Bhaskar D, Pugazhendhi A, Priya TS, Ashok B, Nanthagopal K, Vignesh R, Karthick C. Biodegradable and nonbiodegradable fraction of municipal solid waste for multifaceted applications through a closed loop integrated refinery platform: paving a path towards circular economy. Sci Total Environ. 2020;731:138049. https://doi.org/10.1016/j.scitotenv.2020.138049.
15. Miaoudakis A, et al. Pairing a circular economy and the 5G-enabled internet of things: creating a class of looping smart assets? IEEE Veh Technol Mag. 2020;15(3):20–31. https://doi.org/10.1109/MVT.2020.2991788.

16. Zhang A, Zhong RY, Farooque M, Kang K, Venkatesh VG. Blockchain-based life cycle assessment: an implementation framework and system architecture. Resour Conserv Recycl. 2020;152:104512. https://doi.org/10.1016/j.resconrec.2019.104512.
17. de Oliveira SF, Soares AL. A PLM vision for circular economy. IFIP Adv Inf Commun Technol. 2017;506:591–602. https://doi.org/10.1007/978-3-319-65151-4_52.
18. Plakas G, Ponis ST, Agalianos K, Aretoulaki E. Reverse logistics of end-of-life plastics using industrial IoT and LPWAN technologies – a proposed solution for the bottled water industry. Procedia Manuf. 2020;51:1680–7. https://doi.org/10.1016/j.promfg.2020.10.234.
19. Sassanelli C, Rosa P, Terzi S. Supporting disassembly processes through simulation tools: a systematic literature review with a focus on printed circuit boards. J Manuf Syst. 2021;60:429–48. https://doi.org/10.1016/j.jmsy.2021.07.009.
20. Ekren BY, Mangla SK, Turhanlar EE, Kazancoglu Y, Li G. Lateral inventory share-based models for IoT-enabled E-commerce sustainable food supply networks. Comput Oper Res. 2021;130:105237. https://doi.org/10.1016/j.cor.2021.105237.
21. Chau MQ, Nguyen XP, Huynh TT, Chu VD, Le TH, Nguyen TP, Nguyen DT. Prospects of application of IoT-based advanced technologies in remanufacturing process towards sustainable development and energy-efficient use. Energy Sour Part A Recovery Util Environ Effects. 2021. https://doi.org/10.1080/15567036.2021.1994057.
22. Biagetti G, Crippa P, Falaschetti L, Mansour A, Turchetti C. Energy and performance analysis of lossless compression algorithms for wireless EMG sensors. Sensors. 2021;21:5160.
23. Gulati K, Boddu RSK, Kapila D, Bangare SL, Chandnani N, Saravanan G. A review paper on wireless sensor network techniques in Internet of Things (IoT). Mater Today Proc. 2021;51:161–5.
24. Al-Turjman F, Kamal A, Rehmani MH, Radwan A, Pathan A-SK. The green internet of things (g-iot). Hindawi. 2019;2019:6059343.
25. Bottani E, Manfredi M, Vignali G, Volpi A. Life cycle assessment of RFID implementation in the fresh food supply chain. Int J Real Ther. 2014;6:51–71. https://doi.org/10.3233/RFT-140060.
26. Fazio De et al. Sensors-based treatment system of the organic waste with RFID identification and on-cloud traceability. 2019 IEEE 8th International Workshop on Advances in Sensors and Interfaces (IWASI). 2019; 245–250. https://doi.org/10.1109/IWASI.2019.8791339.
27. El Hadj Youssef W, Abdelli A, Dridi F, Machhout M. Hardware implementation of secure lightweight cryptographic designs for IoT applications. Secur Commun Netw. 2020;2020:8860598.
28. Ngo D-M, Lightbody D, Temko A, Pham-Quoc C, Tran N-T, Murphy CC, Popovici E. HH-NIDS: heterogeneous hardware-based network intrusion detection framework for IoT security. Future Internet. 2023;15:9. https://doi.org/10.3390/fi15010009.
29. Ingemarsdotter E, Jamsin E, Balkenende R. Opportunities and challenges in IoT-enabled circular business model implementation – a case study. Resour Conserv Recycl. 2020;162:105047. https://doi.org/10.1016/j.resconrec.2020.105047.
30. Venkatesh VG, Kang K, Wang B, Zhong RY, Zhang A. System architecture for blockchain based transparency of supply chain social sustainability. Robot Comput Integr Manuf. 2020;63:101896. https://doi.org/10.1016/j.rcim.2019.101896.
31. Feng HH, Wang X, Duan YQ, Zhang J, Zhang XS. Applying blockchain technology to improve agri-food traceability: a review of development methods, benefits and challenges. J Clean Prod. 2020;260:121031. https://doi.org/10.1016/j.jclepro.2020.121031.
32. Vanderroost M, Ragaert P, Verwaeren J, De Meulenaer B, De Baets B, Devlieghere F. The digitization of a food package's life cycle: existing and emerging computer systems in the pre-logistics phase. Comput Ind. 2017;87:15–30. https://doi.org/10.1016/j.compind.2017.01.004.
33. Gligoric N, Krco S, Hakola L, Vehmas K, De S, Moessner K, Jansson K, Polenz I, Van Kranenburg R. Smarttags: IoT product passport for circular economy based on printed sensors and unique item-level identifiers. Sensors. 2019. https://doi.org/10.3390/s19030586.
34. Laskurain-Iturbe I, Arana-Landín G, Landeta-Manzano B, Uriarte-Gallastegi N. Exploring the influence of industry 4.0 technologies on the circular economy. J Clean Prod. 2021;321:128944. https://doi.org/10.1016/j.jclepro.2021.128944.
35. Roy M, Roy A. Nexus of internet of things (IoT) and big data: roadmap for smart management systems (SMgS). IEEE Eng Manag Rev. 2019;47(2):53–65. https://doi.org/10.1109/EMR.2019.2915961.
36 Awan U, Sroufe R, Bozan K. Designing value chains for industry 4.0 and a circular economy: a review of the literature. Sustainability. 2022. https://doi.org/10.3390/su14127084.
37. Abbas MT. Improving the Energy Efficiency of Cellular IoT Device. LICENTIATE THESIS, Karlstad University Studies. 2023;8, ISSN 1403–8099.
38. Boulogeorgos AAA, Diamantoulakis PD, Karagiannidis GK. Low power wide area networks (LPWANs) for Internet of Things (IoT) applications: Research challenges and future trends. arXiv preprint arXiv:1611.07449, 2016.
39. Lavric A, Petrariu AI, Popa V. Long range SigFox communication protocol scalability analysis under large-scale, highdensity conditions. IEEE Access. 2019;7:35816–25.
40. Park J, Salim MM, Jo JH, Sicato JCS, Rathore S, Park JH. CIoTNet: a scalable cognitive IoT based smart city network architecture. Human-centric Comput Inf Sci. 2019;9(1):1–20.
41. Haxhibeqiri J, De Poorter E, Moerman I, Hoebeke J. A survey of LoRaWAN for IoT: from technology to application. Sensors. 2018;18(11):3995.
42. Tin-Yu Wu, Hwang R-H, Vyas A, Lin C-Y, Huang ChiRuei. Persistent periodic uplink scheduling algorithm for massive NB-IoT devices. Sensors. 2022;22(8):2875.
43. Bali MS, Gupta K, Bali KK, Singh PK. Towards energy efficient NB-IoT: A survey on evaluating its suitability for smart applications. Mater Today Proc. 2022;49:3227–34.
44. Kanj M, Savaux V, Le Guen M. A tutorial on NB-IoT physical layer design. IEEE Commun Surv Tutor. 2020;22(4):2408–2446.
45. Sarker KU, Deraman A, Hasan R, Abbas A, Shah B, Ullah A. Monolithic ontological methodology (MOM): an effective software project management approach. J Eng Res (JER). 2022;10(2):118.
46. Sarker KU, Deraman AB, Hasan R, Abbas A. A 4-Layered Plan-driven Model (4LPdM) to Improve Software Development. International Journal of Advanced Computer Science and Applications (IJACSA). 2022; 13(8). https://thesai.org/Downloads/Volume13No8/Paper_68-A_4_Layered_Plan_Driven_Model_4LPdM.pdf.
47. Sarker KU, Saqib M, Hasan R, Mahmood S, Hussain S, Abbas A, Deraman A. A ranking learning model by k-means clustering technique for web scraped movie data. Computers. 2022;11:158. https://doi.org/10.3390/computers11110158.
48. Farahzadi A, Farahsary PS, Rezazadeh J. A survey on IoT fog nano datacenters. Wireless Netw. 2022;28:173–207. https://doi.org/10.1007/s11276-021-02829-2.

49. Singh AK, Swain SR, Lee CN. A metaheuristic virtual machine placement framework toward power efficiency of sustainable cloud environment. Soft Comput. 2023;27:3817–28. https://doi.org/10.1007/s00500-022-07578-8.
50. El-hajj M, Mousawi H, Fadlallah A. Analysis of lightweight cryptographic algorithms on IoT hardware platform. Future Internet. 2023;15:54. https://doi.org/10.3390/fi15020054.
51. Silva C, Cunha VA, Barraca JP, et al. Analysis of the cryptographic algorithms in IoT communications. Inf Syst Front. 2023. https://doi.org/10.1007/s10796-023-10383-9.
52. Dhanda SS, Singh B, Jindal P. Lightweight cryptography: a solution to secure IoT. Wireless Pers Commun. 2020;112(3):1947–80.
53. Fotovvat A, Rahman GM, Vedaei SS, et al. Comparative performance analysis of lightweight cryptography algorithms for IoT sensor nodes. IEEE Internet Things J. 2020;8(10):8279–90.
54. Haque ME, Zobaed S, Islam MU et al. Performance analysis of cryptographic algorithms for selecting better utilization on resource constraint devices. In: 2018 21st International Conference of Computer and Information Technology (ICCIT). 2018; 1–6. https://doi.org/10.1109/ICCITECHN.2018.8631957
55. Hatzivallis G, Fysarakis K, Papaefstathiou I, Manifavas C. A review of lightweight block ciphers. J Cryptogr Eng. 2018;8:141–84.
56. Ngo DM, Temko A, Murphy CC, Popovici E. FPGA Hardware Acceleration Framework for Anomaly-based Intrusion Detection System in IoT. In Proceedings of the 2021 31st International Conference on Field-Programmable Logic and Applications (FPL), Dresden, Germany, 30 August–3 September 2021; 69–75.
57. Bovenzi G, Aceto G, Ciuonzo D, Persico V, Pescapé A. A hierarchical hybrid intrusion detection approach in IoT scenarios. In Proceedings of the GLOBECOM 2020 IEEE Global Communications Conference, Taipei, Taiwan, 7–11 December 2020; 1–7.
58. Kumar, S.; Sahoo, S.; Mahapatra, A.; Swain, A.K.; Mahapatra, K.K. Security enhancements to system on chip devices for IoT perception layer. In Proceedings of the 2017 IEEE International Symposium on Nanoelectronic and Information Systems (iNIS), Bhopal, India, 18–20 December 2017; 151–156.
59. d'Orazio L, Lallet J. Semantic caching framework: An FPGA-based application for IoT security monitoring. Open J Internet Things (OJIoT). 2018;4:150–7.
60. Chéour R, Khriji S, Abid M, Kanoun O. Microcontrollers for IoT: Optimizations, computing paradigms, and uture directions.In Proceedings of the 2020 IEEE 6th World Forum on Internet of Things (WF-IoT), New Orleans, LA, USA, 2–16 June 2020;pp. 1–7.
61. Van Long NH, Lallet J, Casseau E, d'Orazio L. Mascara (ModulAr semantic caching framework) towards FPGA acceleration for IoT security monitoring. In Proceedings of the International Workshop on Very Large Internet of Things (VLIoT 2020), Tokyo, Japan, 4 September 2020.
62. Kalantar A, Zimmerman Z, Brisk P. FA-LAMP: Fpga-accelerated learned approximate matrix profile for time series similarity prediction. In Proceedings of the 2021 IEEE 29th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM), Orlando, FL, USA, 9–12 May 2021; 40–49.
63. Wielgosz M, Karwatowski M. Mapping neural networks to FPGA-based IoT devices for ultra-low latency processing. Sensors. 2019;19:2981.
64. Ioannou L, Fahmy SA. Network intrusion detection using neural networks on FPGA SoCs. In Proceedings of the 2019 29th International Conference on Field Programmable Logic and Applications (FPL), Barcelona, Spain, 9–13 September 2019; 232–238.
65. Yang Z, Liu X, Li T, Wu D, Wang J, Zhao Y, Han H. A systematic literature review of methods and datasets for anomaly-based network intrusion detection. Comput Secur. 2022;116:102675.
66. Hasan M, Islam MM, Zarif MII, Hashem M. Attack and anomaly detection in IoT sensors in IoT sites using machine learning approaches. Internet Things. 2019;7:100059.
67. Alsoufi MA, Razak S, Siraj MM, Nafea I, Ghaleb FA, Saeed F, Nasser M. Anomaly-based intrusion detection systems in IoT using deep learning: a systematic literature review. Appl Sci. 2021;11:8383.
68. Kumar P, Gupta GP, Tripathi R. Design of anomaly-based intrusion detection system using fog computing for IoT network. Autom Control Comput Sci. 2021;55:137–47.
69. Vinayakumar R, Alazab M, Soman K, Poornachandran P, Al-Nemrat A, Venkatraman S. Deep learning approach for intelligent intrusion detection system. IEEE Access. 2019;7:41525–50.
70. Protogerou A, Papadopoulos S, Drosou A, Tzovaras D, Refanidis I. A graph neural network method for distributed anomaly detection in IoT. Evol Syst. 2021;12:19–36.
71. Ullah I, Mahmoud QH. Design and development of RNN anomaly detection model for IoT networks. IEEE Access. 2022;10:62722–50.
72. Dutta V, Choras M, Pawlicki M, Kozik R. A deep learning ensemble for network anomaly and cyber-attack detection. Sensors. 2020;20:4583.
73 Nobakht M, Javidan R, Pourebrahimi A. DEMD-IoT: a deep ensemble model for IoT malware detection using CNNs and network traffic. Evol Syst. 2022. https://doi.org/10.1007/s12530-022-09471-z.
74. Douiba M, Benkirane S, Guezzaz A, Azrour M. An improved anomaly detection model for IoT security using decision tree and gradient boosting. J Supercomput. 2022. https://doi.org/10.1007/s11227-022-04783-y.
75. Thakor VA, Razzaque MA, Khandaker MRA. Lightweight cryptography algorithms for resource-constrained IoT devices: a review, comparison and research opportunities. IEEE Access. 2021;9:28177–93. https://doi.org/10.1109/ACCESS.2021.3052867.
76. Fraga-Lamas P, Lopes SI, Fernandez-Carames TM. Green IoT and edge AI as key technological enablers for a sustainable digital transition towards a smart circular economy: an industry 5.0 Use case. Sensors. 2021. https://doi.org/10.3390/s21175745.
77. Yu Z, Khan S, Mathew M, Umar M, Hassan M, Sajid MJ. Identifying and analyzing the barriers of Internet-of-Things in sustainable supply chain through newly proposed spherical fuzzy geometric mean. Comput Ind Eng. 2022;169:108227. https://doi.org/10.1016/j.cie.2022.108227.
78. Garcia, S.; Parmisano, A.; Erquiaga, M. IoT-23: A Labeled Dataset with Malicious and Benign IoT Network Traffic; Stratosphere Lab.:Praha, Czech Republic, 2020.
79. Hegde M, Kepnang G, Al Mazroei M, Chavis JS, Watkins L. Identification of Botnet Activity in IoT Network Traffic Using Machine Learning. In Proceedings of the 2020 International Conference on Intelligent Data Science Technologies and Applications (IDSTA), Valencia, Spain, 19–22 October 2020; 21–27.
80. Storcheus D, Rostamizadeh A, Kumar S. A survey of modern questions and challenges in feature extraction. In Proceedings of the Feature Extraction: Modern Questions and Challenges. PMLR, Montreal, QC, Canada, 11 December 2015; 1–18.

81. Hussain F, Abbas SG, Fayyaz UU, Shah GA, Toqeer A, Ali A. Towards a Universal Features Set for IoT Botnet Attacks Detection. arXiv 2020, arXiv:2012.00463.
82. Nguyen TD, Marchal S, Miettinen M, Fereidooni H, Asokan N, Sadeghi AR. DÏoT: A federated self-learning anomaly detection system for IoT. In Proceedings of the 2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS), Dallas, TX, USA, 7–9 July 2019; 756–767.
83. Stoian NA. Machine Learning for Anomaly Detection in IoT Networks: Malware Analysis on the IoT-23 Data Set. Bachelor Thesis, University of Twente, Enschede, The Netherlands, 2020.
84. Mothukuri V, Khare P, Parizi RM, Pouriyeh S, Dehghantanha A, Srivastava G. Federated-learning-based anomaly detection for IoT security attacks. IEEE Internet Things J. 2021;9:2545–54.
85. Vaccari I, Chiola G, Aiello M, Mongelli M, Cambiaso E. MQTTset, a new dataset for machine learning techniques on MQTT. Sensors. 2020;20:6578.
86. Manimurugan S, Al-Mutairi S, Aborokbah MM, Chilamkurti N, Ganesan S, Patan R. Effective attack detection in internet of medical things smart environment using a deep belief neural network. IEEE Access. 2020;8:77396–404.
87. Yin C, Zhang S, Wang J, Xiong NN. Anomaly detection based on convolutional recurrent autoencoder for IoT time series. IEEE Trans Syst Man Cybern Syst. 2020;52:112–22.
88. Poonam J, Brahmjit S. Security-performance tradeofs in a class of wireless network scenarios. J Netw Syst Manag. 2017;25(1):83–121.
89. Hamann M, Krause M, Meier W. LIZARD—a lightweight stream cipher for power constrained devices. IACR Transm Symmetric Cryptol. 2017;1:45–79. https://doi.org/10.13154/tosc.v2017.i1.45-79.
90. Dubrova E, Hell M. Espresso: a stream cipher for 5G wireless communication systems. J Cryptogr Commun. 2017;9(2):273–89.
91. Bussi K, Dey D, Kumar M, Dass BK. Neeva: A Lightweight Hash Function. IACR Cryptology ePrint Archive, (042). 2016. https://eprint.iacr.org/2016/042.pdf.
92 Schinianakis D. Alternative security options in the 5G and IoT Era. IEEE Circuits Syst Magzine. 2017. https://doi.org/10.1109/MCAS.2017.2757080.
93. Martínez I, Zalba B, Trillo-Lado R, Blanco T, Cambra D, Casas R. Internet of things (IoT) as sustainable development goals (SDG) enabling technology towards smart readiness indicators (SRI) for university buildings. Sustainability. 2021;13:7647. https://doi.org/10.3390/su13147647.
94. Sarker KU, Yunus F, Deraman A. Penetration taxonomy: a systematic review on the penetration process, framework, standards, tools, and scoring methods. Sustainability. 2023;15:10471. https://doi.org/10.3390/su151310471.
95. Manthiram A, Lutkenhaus JL, Fu Y, Bai P, Kim BG, Lee SW, Okonkwo E, Penner RM. Technological pathways toward sustainable batteries. One Earth. 2022;5(3):203–6. https://doi.org/10.1016/j.oneear.2022.02.010.
96. Wu D, Chen L, Li H, Wu F. Solid-state lithium batteries-from fundamental research to industrial progress. Progress Mater Sci. 2023;139:101182.
97. Turan MS, McKay AK, Çalık C, Chang D, Bassham M. NISTIR 8268: Status Report on the First Round of the NIST Lightweight Cryptography Standardization Process. NIST 2019. https://doi.org/10.6028/NIST.IR.8268
98. Ilkhom B, Khan A, Bakhtiyor A, Das R, Khudoykulov Z. Integral cryptanalysis: a new key determination technique for 3-phase Kuznyechik encryption. Eng Res Express. 2023. https://doi.org/10.1088/2631-8695/ace58f.
99. Das R, Khan A, Arya R, Ilkhom B, Bakhtiyor A, Safoyev N, Khudoykulov Z. SSKA: secure symmetric encryption exploiting Kuznyechik algorithm for trustworthy communication. Int J Syst Assu Eng Manag. 2024;15(6):2391–400. https://doi.org/10.1007/s13198-024-02253-7.
100. Das R, Dash D, Sarkar MK. HTMS: fuzzy based hierarchical trust management scheme in WSN. Wirel Pers Commun. 2020;112:1079–112. https://doi.org/10.1007/s11277-020-07092-w.
101. Das R, Singh M, Majumder K. SGSQoT: A Community-Based Trust Management Scheme in Internet of Things. In: Chakraborty M, Chakrabarti S, Balas V, Mandal J (eds). Proceedings of International Ethical Hacking Conference 2018. Advances in Intelligent Systems and Computing. 2019; 811. Springer, Singapore. https://doi.org/10.1007/978-981-13-1544-2_18
102. Rana M, Mamun Q, Islam R. Lightweight cryptography in IoT networks: a survey. Future Gener Comput Syst. 2022;129:77–89. https://doi.org/10.1016/j.future.2021.11.011.
102 Khan MN, Rao A, Camtepe S. Lightweight cryptographic protocols for IoT-constrained devices: a survey. IEEE Internet Things J. 2021;8(6):4132–56. https://doi.org/10.1109/JIOT.2020.3026493.
104. Singh S, Sharma PK, Moon SY, et al. Advanced lightweight encryption algorithms for IoT devices: survey, challenges and solutions. J Ambient Intell Human Comput. 2024;15:1625–42. https://doi.org/10.1007/s12652-017-0494-4.
104 Amrita, Ekwueme CP, Adam IH, Dwivedi A. Lightweight cryptography for internet of things: a review. EAI Endorsed Trans IoT. 2024. https://doi.org/10.4108/eetiot.5565.
106. Rao V, Prema KV. A review on lightweight cryptography for internet-of-things based applications. J Ambient Intell Human Comput. 2021;12:8835–57. https://doi.org/10.1007/s12652-020-02672-x.